

POORNESHA H M

Bengaluru, India (Open to relocate to Germany)

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Summary

PCB Design Engineer with 14+ years delivering PCB layout and release across automotive, semiconductor, and aerospace programs in Europe, USA, and India. Owns end-to-end ECAD delivery from schematic review and library creation to placement, routing, Gerber/ODB++ release, and supplier DFM/DFA/DFT closure. Strong in SI-aware high-speed layout, EMI/EMC-driven design practices, and PLM-controlled documentation using Teamcenter and Jira.

Core Skills

- **PCB Design & Layout:** Placement, Routing, Multilayer (up to 24 layers), Gerber, ODB++
- **Library & Schematic:** PCB Library Creation, Footprints, Symbols, Schematic Drawing, ORCAD
- **Power & Safety:** 48V–650V, Chargers, DC-DC, PDU, DBC Modules, Creepage, Clearance
- **SI / EMI EMC:** Pre/Post-Layout SI, EMI/EMC Guidelines, Basic PI (Voltas), Constraint Rules
- **DFX & Quality:** PCB Gerber Verify, DFM/DFA/DFT, Quality Assurance, Vendor Feedback Closure
- **Management & Soft Skills:** Team Management, Talent Acquisition, Communication, Problem Solving

Experience

Tata Electronics

November 2024 - Present

Manager PCB Design & Development

Bangalore, India

- Lead end-to-end PCB design delivery for automotive and semiconductor programs for global customers.
- Own SI-aware layout using stackup planning, constraints, and multi-layer routing with Cadence tools.
- Deliver boards meeting performance and manufacturability targets; improved first-pass builds by 15%.
- Manage SOWs, schedules, vendor DFM/DFA/DFT reviews, and mentored 3+ PCB design engineers.

Liebherr Electronics Drive GmbH

April 2024 - November 2024

Lead PCB Design

Lindau, Germany

- Owned high-speed PCB layout up to 24 layers for embedded and industrial platforms in Germany.
- Routed 1-10 GHz links: Ethernet, USB 3.0, PCIe Gen1-Gen5, DDR3/4/5, SERDES, WiFi, Bluetooth.
- Accelerated releases by 20% via clean ODB++, Gerber, BOM, and fab/assy drawings using Teamcenter.
- Reduced SI rework by 15% through schematic collaboration and pre/post-layout integrity reviews.

Bosch Global Software Technologies

May 2021 - April 2024

Lead PCB Design

Bengaluru, India

- Designed power electronics PCBs for 48V–650V systems including chargers, DC-DC, and PDU boards.
- Supported DBC substrate modules and emergency vehicle systems with robust power layout practices.
- Improved manufacturability by closing DFA/DFT/DFM feedback and updating release checklists.
- Enabled faster integration via ECAD-MCAD alignment and clear documentation for 10+ stakeholders.

Mahindra Electric Limited

October 2019 - May 2021

Assistant Manager PCB Design

Bengaluru, India

- Owned PCB layout delivery for power electronics and dashboard cluster modules in EV programs.
- Executed ECAD-MCAD integration with cross-functional teams and resolved mechanical constraints early.
- Improved SI outcomes by running initial checks and driving issue closure with senior analysis support.
- Reduced layout rework by 15% through schematic collaboration and review-driven constraint updates.

Ather Energy Limited

May 2016 - September 2019

Executive PCB Design

Bangalore, India

- Delivered schematic entry, footprint creation, placement, and routing for vehicle electronics PCBs.
- Produced Gerber files and PCB documentation and coordinated releases with vendors and stakeholders.
- Accelerated vendor turnarounds by 20% by closing DFM/DFA/DFT feedback and clarifying requirements.
- Applied IPC 610, IPC-D-325, and IPC 2221/2222 guidance to strengthen assembly and design compliance.

Bit Mapper Technologies

September 2014 - August 2015

CAD PCB Design

Pune, India

- Owned component placement, schematic capture, and multi-layer PCB routing for customer designs.
- Created IPC and ANSI-aligned footprints and symbols and validated libraries before release.
- Improved manufacturability and cost outcomes through routing discipline and vendor-ready outputs.
- Delivered fab and assembly drawings and kept documentation complete across 10+ design releases.

NCS

October 2012 - February 2014

Junior CAD/CAM Engineer

Bangalore, India

- Supported schematic entry and PCB layout in ORCAD and executed routing updates under library governance.
- Built footprints and managed component libraries per IPC 7351B, improving reuse across 30+ parts.
- Reduced CAM rework by 15% by generating clean Gerber outputs and documentation for manufacturing.

Education

BMS College of Engineering (VTU University)

2015 - 2020

Bachelor of Engineering (B.E) in Electronics & Communication

Bengaluru, India

PES University

2008 - 2011

Diploma in Electronics & Communication Engineering

Bengaluru, India

Languages

- English - Full Professional Proficiency (C2)
- German - Intermediate Working Proficiency (B1)